PROJECT REPORT

E-Commerce Sales Data Exploration, Customer Segmentation & Strategic Recommendations

Author: Harsh Kumar Verma | Date: August 7, 2026

---

1. Document Overview & Project Structure

This report integrates findings from a multi-stage analysis using SQLite, Python, and Power BI. By examining sales patterns, product performance, and regional trends, we have developed a clear picture of customer behavior and identified actionable strategic opportunities. The project infrastructure is summarized below:

S.No.	Asset Name	Description & Scope
1	Dataset	Original raw dataset file containing order-level transactions.
2	SQLite Script	SQL script file utilized for initial data exploration, schema understanding, and relational checks.
3	Python Scripts	Comprehensive cleaning, transformation, statistical analysis, and data visualization code.
4	Cleaned Dataframe	Transformed CSV file extracted post-cleaning and preprocessing in Python for exploratory analysis.
5	Sales Dashboards	Power BI dashboards showcasing interactive customer behavior, trends, and regional insights.
6	Final Report	Executive summary containing core findings, methodology, statistical verification, and strategic recommendations.

2. Data Exploration & Initial Validation

ORDERNUMBER vs. ORDERLINENUMBER Relationship

A sample audit of order numbers against line items confirms that each primary order typically bundles multiple products, providing a granular view of transaction health.

Data Completeness & Anomaly Resolution

The 2005 dataset concludes after the second quarter. While this creates an apparent drop in annual performance, it simply reflects the cutoff point for data collection. Geographic standardization efforts also revealed that initial records incorrectly treated Japan as a standalone territory; these entries were subsequently reclassified under the APAC territory to maintain regional accuracy. Finally, the territory code "NA" for North America was initially misinterpreted by pandas as a null value, and converting these strings corrected the non-null count from 1,749 to the full 2,823 records.

3. Power BI Insights & Customer Behavioral Segmentation

Order Value Trends & Category Performance

Analysis of order value trends reveals a significant Average Order Value peak in February 2003, reaching $47,000. Although transaction volume was relatively low during this period, these high-value deals highlight the substantial influence of our top-tier accounts. From a product perspective, the "Cars" category remains our most resilient asset, distinguishing itself as the only product line sold every single month of the tracking period. Motorcycles and Trucks & Buses follow as steady secondary contributors to overall revenue.

> **RECOMMENDATION 1: Product Line & Bundle Expansion**
>
> *Focus resources on 'Cars,' the most resilient category. Use the top-selling item 'S18_3232' as a bundle anchor to stimulate interest in underperforming lines like 'Trains'.*

Customer Classification by Value & Frequency

By mapping customers against order frequency (threshold: 14) and total lifetime value (threshold: $100,000), we identified distinct engagement tiers:

Customer Segment	Order Frequency	Monetary Value	Customer Count
Low-Frequency & Low-Value	< 14 Orders	< $100,000	54 Customers (58.7%)
High-Frequency & High-Value	≥ 14 Orders	≥ $100,000	2 Customers (2.2%)

Approximately 97.8% of the customer base is low-frequency. Increasing repeat purchase rates is the primary lever for sustainable growth.

> **RECOMMENDATION 2: Retention & Re-Order Incentive Program**
>
> *Shift the low-frequency buyer segment toward repeat activity by implementing bounce-back discounts on subsequent orders and a points-based loyalty program to incentivize long-term retention.*

4. Python Seasonal & Regional Analysis

Seasonal Demand Patterns

Time-series analysis highlights a massive demand surge every November, which consistently leads in both revenue and volume, significantly outperforming other quarters. This surge aligns with the global holiday retail cycle and pre-Christmas clearance campaigns, confirming that the business is heavily influenced by year-end seasonal spending patterns.

> **RECOMMENDATION 3: Year-Round Marketing Campaigns**
>
> *Balance annual revenue distribution by scheduling mid-year promotional events, such as Summer Clearance, rather than relying exclusively on Q4 surges.*

Geographic Distribution & Territory Breakdown

The United States remains the primary driver of global order volume, followed by strong performances in the French and Spanish markets. However, demand varies significantly by territory, suggesting that a localized inventory strategy would be more efficient than a universal approach.

Territory	Dominant / Key Category	Sales Contribution & Insights
North America (NA)	Motorcycles	Generates 45% of total global sales in the Motorcycles category.
EMEA	Trains	Accounts for 61% of total global sales in the Trains category.
APAC	Trucks & Buses	Overall weak total sales share; highest category contribution is 16% in Trucks & Buses. Zero orders recorded in multiple months (e.g., Jan, Feb, Jun, Aug, Dec 2003; Mar, May, Oct 2004).

> **RECOMMENDATION 4: APAC Market Activation**
>
> *Drive APAC market activation by applying the loyalty and bundling strategies defined above to build consistent purchasing habits.*

5. SQL Query Exploration Findings

Product code S18_3232 is the highest-volume item, with 1,774 units across 52 transactions. 'Classic Cars' dominates, moving over 33,000 units. Medium-sized transactions provide the most stable revenue.

Regional analysis reveals that despite high performance in the US, volume in Ireland remains low, suggesting a need for targeted localized marketing. Conversely, specific pockets of high demand, such as "Cars" in California and "Classic Cars" throughout EMEA, indicate strong regional brand loyalty that can be further leveraged.

6. Statistical Overview & Deal Size Metrics

Transaction Statistics & Max Order Profile

The mean sales value per order line currently stands at $3,553.88, which serves as a baseline for performance expectations. Our largest single line-item record reached $14,082.80 in April 2005, attributed to "The Sharp Gifts Warehouse" as part of a significant 12-item order focused entirely on the "Cars" category.

Deal Size Segmentation Thresholds

Order line transaction sizes are categorized according to monetary values:

Deal Size Category	Monetary Threshold (USD)	Description & Highlights
Small	< $3,000	Low-value transaction lines; highest overall transaction volume.
Medium	$3,000 – $7,000	Mid-tier transaction lines; forms the majority of total business revenue.
Large	> $7,000	High-value transaction lines; repeated most frequently by top 3 enterprise accounts.

Key Customer Accounts & City Order Extremes

Three enterprise pillars consistently drive our "Large" deal volume: Euro Shopping Channel, Mini Gifts Distributors Ltd, and The Sharp Gifts Warehouse; maintaining these accounts is essential. Regarding city-level benchmarks, Madrid stands out as our most active city with 31 orders, while Munich recorded only a single order, highlighting a clear area for future market development.

7. Conclusion & Strategic Summary

This project synthesized SQL, Python, and Power BI analyses to evaluate sales trends, product performance, and customer behavior. Standardizing territory data and resolving missing value parsing established a clean foundation of 2,823 records for reliable decision-making.

Key Findings & Actionable Roadmap

To drive growth, the company should leverage the high-performing "Cars" category and use the top-selling item S18_3232 as an anchor for promotional bundles to boost categories like "Trains." Additionally, implementing targeted loyalty programs and re-purchase incentives is critical for converting the 97% of low-frequency buyers into repeat customers.

This analysis confirms that while Q4 performance remains strong, seasonal volatility and low repeat-purchase rates limit consistent revenue. By leveraging top-performing product categories like 'Cars' for bundling and expanding loyalty programs into underperforming regions like APAC and Ireland, the business can stabilize cash flow and reduce reliance on end-of-year sales spikes.